Fake News Detection

Group: Group 2B

1. Introduction

The digital age is characterized by the rapid and widespread dissemination of information. The dissemination of news articles with implausible, dubious, dishonest or unreliable content propagates misinformation, erodes public trust, and manipulates public opinion. Manual fact-checking is unfeasible and unviable owing to the sheer volume of "informational textual content" posted across news dissemination websites and shared across social media platforms. This project presents a robust machine learning framework designed to automatically detect fake news by classifying news articles as either genuine (Class 1) or fabricated (Class 0). The methodology centers on natural language processing (NLP), Logistic Regression, Ensemble models and Deep Learning to analyze the linguistic features, sentiment, and contextual features in text data.

1.1 Business Understanding

In today's information-driven society, the spread of misinformation and fake news through online news dissemination platforms and social media platforms has become a serious threat to public trust, democracy, health communication, and media integrity. The ability to automatically distinguish fake from real news is essential for:

- Social media platforms (e.g., Meta, Twitter/X, Reddit).
- Fact-checking organizations.
- News aggregators (e.g., Google News, Apple News).
- The general public.

1.2 Problem Statement:

The proliferation of fake news on digital platforms poses a significant risk to public trust and decision-making. Traditional methods of identifying fake news are slow and resource-intensive, making it difficult to manage the vast volume of online content.

1.3 Proposed solution

This project aims to develop an automated binary classification system using transformer-based Natural Language Processing (NLP) models to detect and flag fake news articles with high accuracy. By leveraging advanced NLP techniques, the system will offer a scalable solution to help identify harmful misinformation, improving information credibility and supporting the fight against digital manipulation.

1.4 Objectives

1. Perform Extraploratory Data Analysis
2. Build a Baseline Logistic Regression Model to predict class of an article.

3. Build an XGBoost model, a LSTM neural network and finetuned the RoBERTa-base Transformer.
4. Interpret best performing ML model using the LIME library.
5. Deploy selected model using FastAPI and Streamlit.

1.5 Success Criteria

This project aims to catch the most fake news articles. However, the classes for the target variable of the dataset used in this project is imbalanced. Thus, the success criteria for this project is the ***macro-averaged recall*** (the average recall across all classes, treating each class equally). Macro-averaged recall is particularly important when dealing with imbalanced datasets because it ensures that performance on the minority class is not overshadowed by the majority class. We aim to achieve a ***macro-averaged recall*** of over **76%** for the deployed

2. Data Loading, Feature Engineering, and Preprocessing

2.1 Data Loading

In [1]:
```
# Declare dataset path
import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
/kaggle/input/news-article-credibility-detection-dataset/data.csv
```

In [2]:
```
# Declare images directory
import os
images_dir = '/kaggle/working/images'
os.makedirs(images_dir, exist_ok=True)
print(os.listdir('/kaggle/working/'))
```

```
['images', '.virtual_documents']
```

In [4]:
```
# !pip install -U scikit-learn==1.5.2 imbalanced-learn==0.12.3 xgb
```

In [6]:

```
# Import basic libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from wordcloud import WordCloud
from collections import Counter
import io

# Import NLP tools and NLTK library modules
import nltk
from nltk.corpus import stopwords, wordnet
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.stem import WordNetLemmatizer, PorterStemmer
from nltk import pos_tag
from nltk.util import bigrams
from nltk.util import trigrams
import string
import re
import spacy

# Import scikit-learn library's classes, tools, and modules
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split, cross_val_sc
from sklearn.feature_extraction.text import CountVectorizer, TfidfV
from sklearn.metrics import accuracy_score, classification_report,
from sklearn.metrics import f1_score, precision_score, recall_score
from sklearn.linear_model import LogisticRegression
from xgboost import XGBClassifier
from sklearn.utils.class_weight import compute_class_weight
from imblearn.over_sampling import SMOTE
from imblearn.under_sampling import RandomUnderSampler
import joblib
from gensim.models import Word2Vec

# Import tensorflow, and keras modules
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Model, load_model
from tensorflow.keras.layers import TextVectorization, Embedding, C
from tensorflow.keras.layers import Dropout, Dense, LSTM, Input, Ma
from tensorflow.keras.metrics import Precision, Recall, AUC
from transformers import TFRobertaForSequenceClassification, Robert
from transformers import logging as transformers_logging

# Import transformers and pytorch modules
from transformers import RobertaTokenizerFast, RobertaForSequenceCl
from datasets import Dataset
import torch
from torch.utils.data import Dataset
import os

# Import Model Interpretation Libraries and modules
from lime.lime_text import LimeTextExplainer
import matplotlib.patches as mpatches
```

```
# Disable warnings
import warnings
warnings.filterwarnings('ignore')
```

In [7]:
```
path = '/kaggle/input/news-article-credibility-detection-dataset/da
data = pd.read_csv(path)
data.head()
```

Out[7]:

	news_url	title	extracted_article_text	news_type
0	https://www.bustle.com/p/will-the-royals-retur...	Will 'The Royals' Return For Season 5? This St...	Entertainment With a royal wedding and a coup ...	gossip
1	https://www.foxnews.com/entertainment/naya-riv...	Naya Rivera refiles for divorce from Ryan Dors...	This material may not be published, broadcast,...	gossip
2	https://www.unitedbypop.com/style/fashion/tayl...	Outfit ideas for Taylor Swift's Reputation Tour	United By Pop - United Kingdom. United States....	gossip
3	https://variety.com/2018/music/news/scott-hutc...	Scott Hutchison Dead: Frightened Rabbit Frontm...	By Robert Mitchell A body found Thursday night...	gossip
4	https://www.etonline.com/kate-middleton-gives-...	Kate Middleton Gives Birth to Royal Baby No. 3...	Baby No. 3 is officially here! Kate Middleton ...	gossip

In [8]:
```
# Check shape
data.shape
```

Out[8]:
```
(16044, 5)
```

In [9]:
```
# Check duplicates
data.duplicated().sum()
```

Out[9]:
```
928
```

In [10]:
```
# Drop duplicates
data.drop_duplicates(inplace=True)
```

In [14]:
```
# Confirm no duplicates
data.duplicated().sum()
```

Out[14]:
```
0
```

In [17]:
```
# Inspect column attributes to check missing values
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 15116 entries, 0 to 16043
Data columns (total 5 columns):
 #   Column                 Non-Null Count  Dtype 
---  ------                 --------------  ----- 
 0   news_url               15116 non-null  object
 1   title                  15116 non-null  object
 2   extracted_article_text  15116 non-null  object
 3   news_type              15116 non-null  object
 4   class                  15116 non-null  int64 
dtypes: int64(1), object(4)
memory usage: 708.6+ KB
```

- Data was successfully loaded and duplicates removed.
- There are no entries with missing values

2.2 Feature Engineering

In [18]:
```
# Engineer the domain feature
from urllib.parse import urlparse
data['domain'] = data['news_url'].apply(lambda x: urlparse(x).netlo

# Engineer the article feature
data['article'] = data['title'] + " " + data['extracted_article_tex

# Engineer the text length
data['num_sentences']=data['extracted_article_text'].apply(lambda x

# Engineer the text length
data['text_length']=data['extracted_article_text'].apply(lambda x:
data.head(2)
```

Out[18]:

	news_url	title	extracted_article_text	news_type	cla
0	https://www.bustle.com/p/will-the-royals-retur...	Will 'The Royals' Return For Season 5? This St...	Entertainment With a royal wedding and a coup ...	gossip	
1	https://www.foxnews.com/entertainment/naya-riv...	Naya Rivera refiles for divorce from Ryan Dors...	This material may not be published, broadcast,...	gossip	

In [19]:
```
data['domain'].value_counts().nlargest(5)
```

Out[19]:
```
domain
missing                3237
people.com             1224
www.dailymail.co.uk     682
www.etonline.com        536
www.usmagazine.com      512
Name: count, dtype: int64
```

2.3 Data Preprocessing

- Define a function for ***text cleaning, normalization, word tokenizing, Part-of-Speech tagging, lemmatization*** and ***removing stopwords***, using the **spaCy** library.

In [20]:

```
# Load spaCy model
nlp = spacy.load('en_core_web_sm', disable=['parser', 'ner'])
# Define function to clean text using spaCy
def cleaned_text(text, debug=False):
    # Convert lists to strings if necessary
    if isinstance(text, list):
        text = " ".join(text)
    # Ensure text is a string
    if not isinstance(text, str):
        text = str(text)
    text = text.lower()# Lowercase all characters
    text = re.sub(r'https?://\S+|www\.\S+', '', text) # Remove URL
    text = re.sub(r'\[.*?\]', '', text) # Remove content enclosed
    text = re.sub(r'<.*?>+', '', text)  # Remove content enclosed
    text = re.sub(r'\n', ' ', text)     # Replace newline characte

    # Define chunk size
    chunk_size = 900000
    cleaned_tokens = []
    # Process text in chunks if it's too long
    if len(text) > chunk_size:
        chunks = [text[i:i + chunk_size] for i in range(0, len(tex
    else:
        chunks = [text]

    for chunk in chunks:
        doc = nlp(chunk)
        # Extract lemmatized tokens, excluding stopwords,and punctu
        for token in doc:
            if (not token.is_stop and
                not token.is_punct):
                #not token.is_digit and
                #len(token.lemma_) > 2):
                # Remove any remaining non-alphabetic characters f
                cleaned = re.sub(r'[^a-zA-Z]', '', token.lemma_)
                if cleaned:  # Only include non-empty strings
                    cleaned_tokens.append(cleaned)
    return cleaned_tokens

# Apply the clean_text function to the 'spacy' column
data['spacy'] = data['article'].apply(cleaned_text)
```

In [21]:
```
# Join tokens
data['text'] = data['spacy'].apply(lambda x: ' '.join(x))
data.head(2)
```

Out[21]:

	news_url	title	extracted_article_text	news_type	cla
0	https://www.bustle.com/p/will-the-royals-retur...	Will 'The Royals' Return For Season 5? This St...	Entertainment With a royal wedding and a coup ...	gossip	
1	https://www.foxnews.com/entertainment/naya-riv...	Naya Rivera refiles for divorce from Ryan Dors...	This material may not be published, broadcast,...	gossip	

In [22]:
```
# Preview a random sample
data.iloc[99, 10]
```

Out[22]:
```
'blac chyna cozy hot felon jeremy meek pic look like blac chyna h
ot felon jeremy meek work chyna share picture posing arm snapchat
wednesday night show tattoo chyna don skintight lace orange dress
meek keep casual camouflage print shirt rip jean news early snapc
hat post chyna appear photo shoot lead speculate possibly star ca
mpaign meek model attractive mugshot go viral headline summer rel
ationship topshop heiress chloe green meek photograph kiss green
luxury yacht coast turkey july married wife melissa time shortly
picture surface meek file separation melissa year marriage news m
eek green shy pack pda watch video'
```

In [23]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 15116 entries, 0 to 16043
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   news_url                15116 non-null  object
 1   title                   15116 non-null  object
 2   extracted_article_text  15116 non-null  object
 3   news_type               15116 non-null  object
 4   class                   15116 non-null  int64
 5   domain                  15116 non-null  object
 6   article                 15116 non-null  object
 7   num_sentences           15116 non-null  int64
 8   text_length             15116 non-null  int64
 9   spacy                   15116 non-null  object
 10  text                    15116 non-null  object
dtypes: int64(3), object(8)
memory usage: 1.4+ MB
```

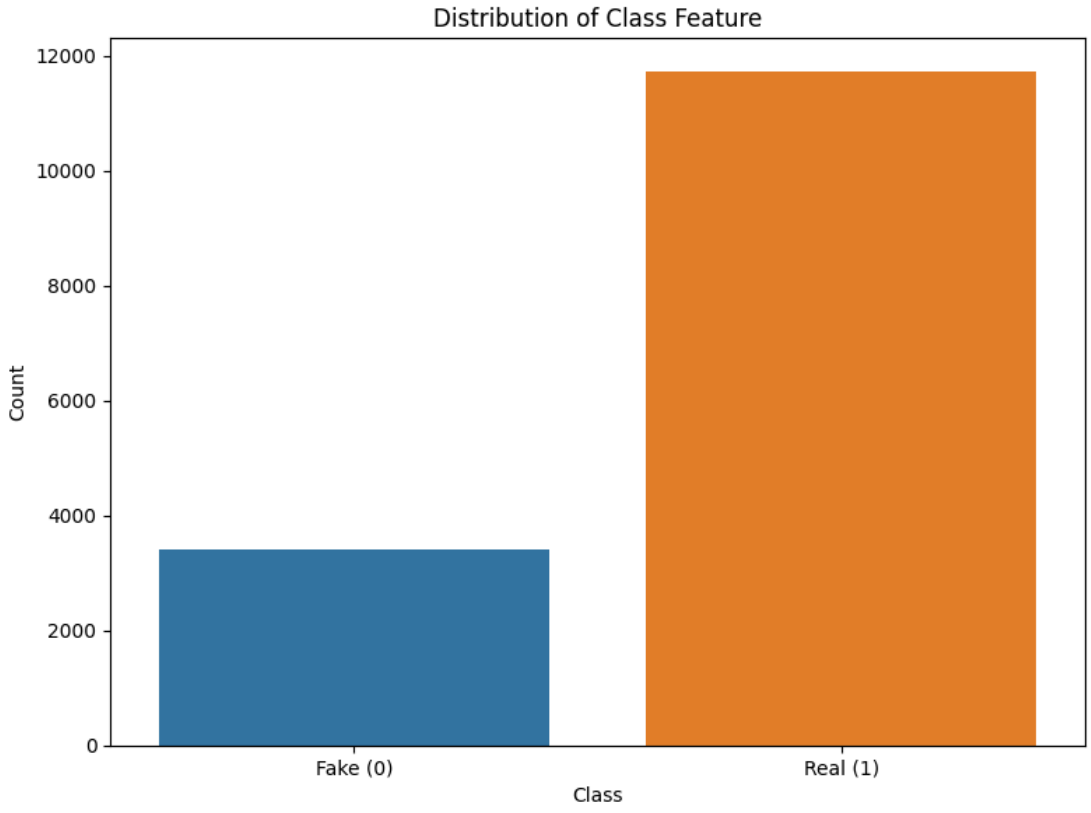
Objective 1: Perform Extraploratory Data Analysis

Target Variable Class Distributions

In [24]:
```
# Print value counts for each class
data['class'].value_counts()
```

Out[24]:
```
class
1    11717
0     3399
Name: count, dtype: int64
```

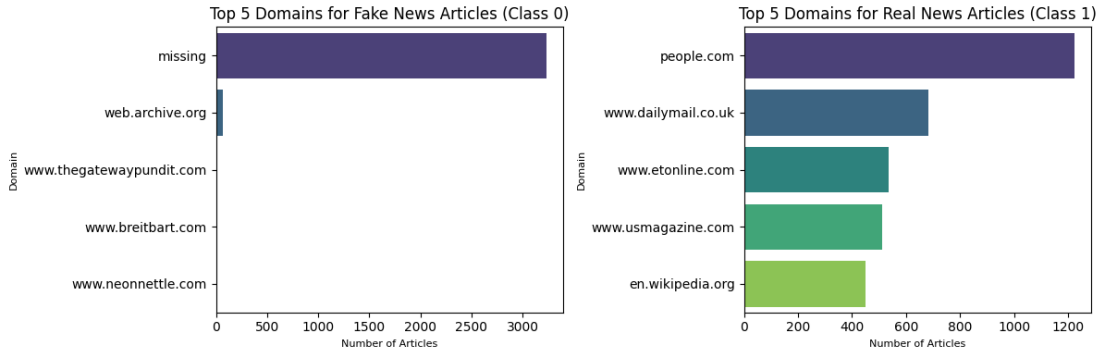
In [25]:
```
# Visualize Class Distribuction
plt.figure(figsize=(8, 6))
sns.barplot(x='class', y='class', data=data, estimator=len)
plt.xlabel('Class')
plt.ylabel('Count')
plt.title('Distribution of Class Feature')
plt.xticks([0, 1], ['Fake (0)', 'Real (1)'])
plt.tight_layout()
plt.savefig('/kaggle/working/images/class-distributions.png', dpi=
plt.show()
```



Top-5 Domains for Fake and Real News Articles

In [26]:

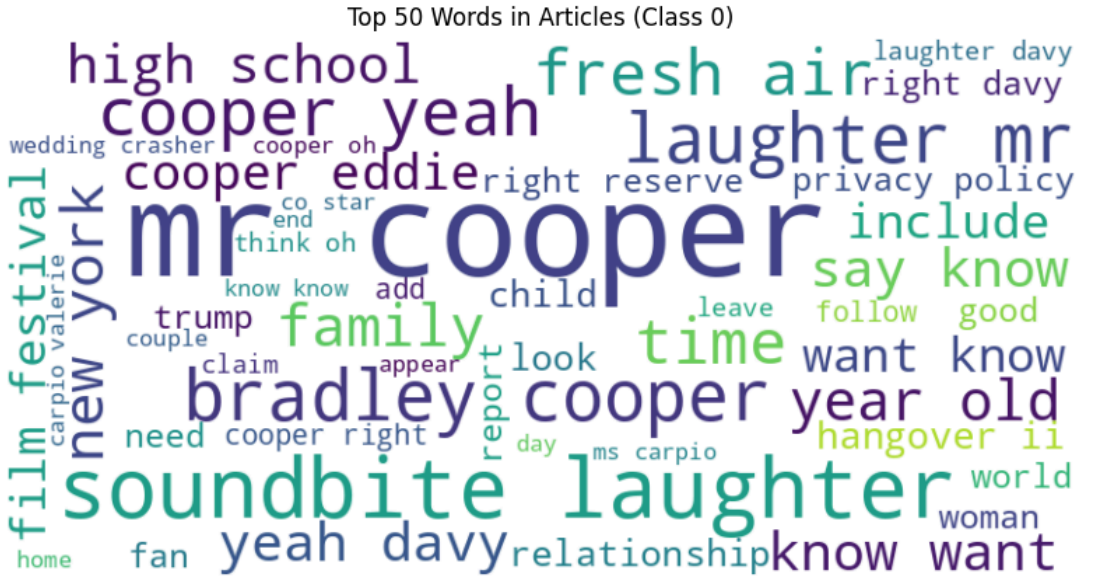
```
# Visualize top 5 domains for fake news and real news
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4))

# Plot for class 0 (Fake News)
fake_data = data[data['class'] == 0]
domain_counts = fake_data['domain'].value_counts()
top_5_domains = domain_counts.nlargest(5)
sns.barplot(x=top_5_domains.values, y=top_5_domains.index, palette=
ax1.set_title('Top 5 Domains for Fake News Articles (Class 0)', for
ax1.set_xlabel('Number of Articles', fontsize=8)
ax1.set_ylabel('Domain', fontsize=8)

# Plot for class 1 (Real News)
fake_data = data[data['class'] == 1]
domain_counts = fake_data['domain'].value_counts()
top_5_domains = domain_counts.nlargest(5)
sns.barplot(x=top_5_domains.values, y=top_5_domains.index, palette=
ax2.set_title('Top 5 Domains for Real News Articles (Class 1)', for
ax2.set_xlabel('Number of Articles', fontsize=8)
ax2.set_ylabel('Domain', fontsize=8)
plt.tight_layout()
plt.savefig('/kaggle/working/images/top-5-domains-for-fake-and-real
plt.show()
```



Most Frequent/ Significant Words Among Fake News Articles (Label 0)

In [27]:

```
# Filter data for class 0
class_0_data = data[data['class'] == 0]

# Combine all text from extracted_article_text
text = ' '.join(class_0_data['text'].dropna().astype(str))
wordcloud = WordCloud(width=800, height=400, background_color='whit
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Top 50 Words in Articles (Class 0)')
plt.savefig('/kaggle/working/images/top-50-words-for-class0-wordclo
plt.show()
```



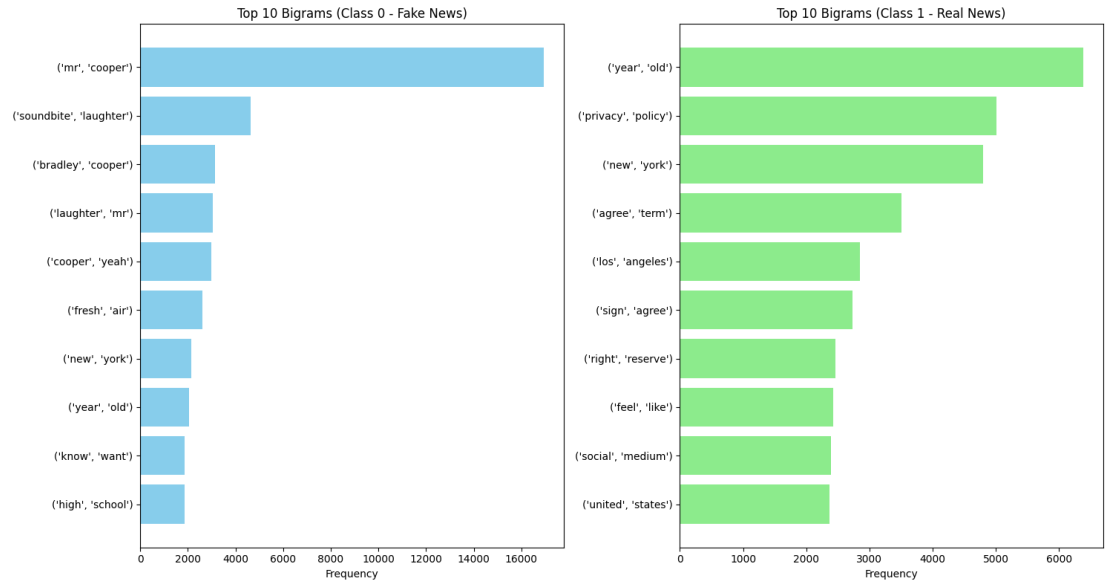
Most Frequent/ Significant Words Among Real News Articles (Label 1)

In [28]:
```
# Filter data for class 1
class_0_data = data[data['class'] == 1]

# Combine all text from extracted_article_text
text = ' '.join(class_0_data['text'].dropna().astype(str))
wordcloud = WordCloud(width=800, height=400, background_color='whit
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Top 50 Words in Articles (Class 1)')
plt.savefig('/kaggle/working/images/top-50-words-for-class1-wordclo
plt.show()
```

Top-10 Bigrams

In [29]:

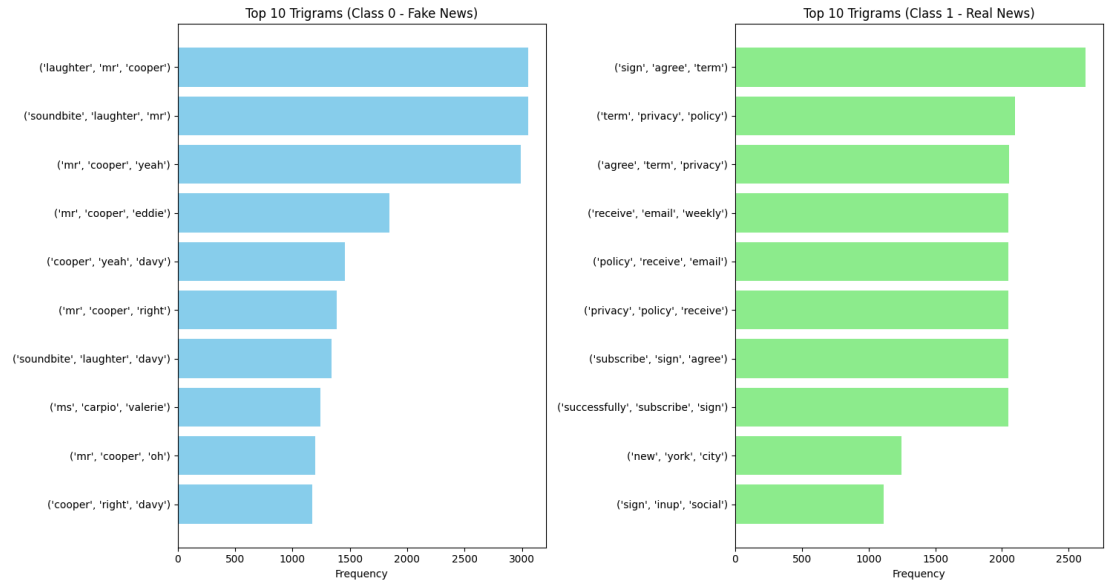
```
# Filter and plot top-10 bigrams for both fake and real news artic
def get_top_bigrams(text_series):
    all_bigrams = []
    for text in text_series.dropna():
        tokens = word_tokenize(text.lower())
        bigram_list = list(bigrams(tokens))
        all_bigrams.extend(bigram_list)
    return Counter(all_bigrams).most_common(10)

class_0_data = data[data['class'] == 0]['text']
class_1_data = data[data['class'] == 1]['text']

# Get top 10 bigrams for each class
top_bigrams_class_0 = get_top_bigrams(class_0_data)
top_bigrams_class_1 = get_top_bigrams(class_1_data)
bigrams_0, counts_0 = zip(*sorted(top_bigrams_class_0, key=lambda x
bigrams_1, counts_1 = zip(*sorted(top_bigrams_class_1, key=lambda x

# Create subplots
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 8))
# Plot for class 0
ax1.barh([str(bg) for bg in bigrams_0], counts_0, color='skyblue')
ax1.set_title('Top 10 Bigrams (Class 0 - Fake News)')
ax1.set_xlabel('Frequency')
# Plot for class 1
ax2.barh([str(bg) for bg in bigrams_1], counts_1, color='lightgreer
ax2.set_title('Top 10 Bigrams (Class 1 - Real News)')
ax2.set_xlabel('Frequency')
plt.tight_layout()
plt.savefig('/kaggle/working/images/top-10-bigrams.png', dpi=300, b
plt.show()
```



Top-10 Trigrams

In [30]:

```
# Filter and plot top-10 trigrams for both fake and real news artic
def get_top_trigrams(text_series):
    all_trigrams = []
    for text in text_series.dropna():
        tokens = word_tokenize(text.lower())
        trigram_list = list(trigrams(tokens))
        all_trigrams.extend(trigram_list)
    return Counter(all_trigrams).most_common(10)

# Filter data for class 0 and class 1
class_0_data = data[data['class'] == 0]['text']
class_1_data = data[data['class'] == 1]['text']
top_trigrams_class_0 = get_top_trigrams(class_0_data)
top_trigrams_class_1 = get_top_trigrams(class_1_data)
trigrams_0, counts_0 = zip(*sorted(top_trigrams_class_0, key=lambda
trigrams_1, counts_1 = zip(*sorted(top_trigrams_class_1, key=lambda

# Create subplots
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 8))
# Plot for class 0
ax1.barh([str(tg) for tg in trigrams_0], counts_0, color='skyblue')
ax1.set_title('Top 10 Trigrams (Class 0 - Fake News)')
ax1.set_xlabel('Frequency')
# Plot for class 1
ax2.barh([str(tg) for tg in trigrams_1], counts_1, color='lightgree
ax2.set_title('Top 10 Trigrams (Class 1 - Real News)')
ax2.set_xlabel('Frequency')
plt.tight_layout()
plt.savefig('/kaggle/working/images/top-10-trigrams.png', dpi=300,
plt.show()
```



3. Modelling

In [110]:
```
# Create a copy of data
df = data.copy()

# Define endog and exog
endog = df["class"]
exog = df[['text']]

# First train-test split (70-30)
X_temp, X_test, y_temp, y_test = train_test_split(exog, endog, test

# Second split temp into train and validation (70-30 of remaining
X_train, X_val, y_train, y_val = train_test_split(X_temp, y_temp, t
```

In [111]:
```
# Tokenize text feature for Word2Vec
def tokenize_text(text_series):
    return [word_tokenize(text.lower()) for text in text_series.dro
tokenized_spacy_text = tokenize_text(exog['text'])

# Train Word2Vec model on the tokenized text
word2vec_model = Word2Vec(sentences=tokenized_spacy_text, vector_si

# Function to get average Word2Vec vector for a text
def get_word2vec_vector(text, model, vector_size=100):
    words = word_tokenize(text.lower())
    word_vectors = [model.wv[word] for word in words if word in mod
    if len(word_vectors) == 0:
        return np.zeros(vector_size)
    return np.mean(word_vectors, axis=0)

# Vectorize spacy_text using Word2Vec
X_train_1 = np.array([get_word2vec_vector(text, word2vec_model) for
X_val_1 = np.array([get_word2vec_vector(text, word2vec_model) for t
X_test_1 = np.array([get_word2vec_vector(text, word2vec_model) for

# Check shapes
print("X_train shape:", X_train_1.shape)
print("X_val shape:", X_val_1.shape)
print("X_test shape:", X_test_1.shape)
```

```
X_train shape: (7406, 100)
X_val shape: (3175, 100)
X_test shape: (4535, 100)
```

Objective 2: Build a Baseline Model (Logistic Regression)

- The target variable's classes are imbalanced. Two strategies to address class imbalance are incoporated in the modelling pipelines.
  - **SMOTE** is implemented in a pipeline to generate synthentic samples for the minority class.
  - **Random Undersampling** is implemented in a pipeline to reduce entities for the majority class.

In [116]:

```
# Prepare training and validation sets
X_train_logistic = np.vstack((X_train_1, X_val_1))
y_train_logistic = np.hstack((y_train, y_val))

# Define pipeline for fitting model with imbalanced data
lr_model_imbalanced = LogisticRegression(max_iter=1000, random_stat
cv_scores_imbalanced = cross_val_score(lr_model_imbalanced, X_train
print("\n[Pipeline 1: Imbalanced]")
print("Unbalanced X_train_logistic shape:", X_train_logistic.shape)
print("CV Scores:", cv_scores_imbalanced)
print("Average recall:", cv_scores_imbalanced.mean())

# Define pipeline for fitting model with oversampled minority class
smote = SMOTE(sampling_strategy='minority', random_state=42)
X_train_smote, y_train_smote = smote.fit_resample(X_train_logistic,
lr_model_smote = LogisticRegression(max_iter=1000, random_state=42)
cv_scores_smote = cross_val_score(lr_model_smote, X_train_smote, y_
print("\n[Pipeline 2: SMOTE Oversampling]")
print("SMOTE-balanced X_train_logistic shape:", X_train_smote.shape
print("CV Scores:", cv_scores_smote)
print("Average recall:", cv_scores_smote.mean())

# Define pipeline for fitting model with undersampled majority class
rus = RandomUnderSampler(sampling_strategy='majority', random_state
X_train_rus, y_train_rus = rus.fit_resample(X_train_logistic, y_tra
lr_model_rus = LogisticRegression(max_iter=1000, random_state=42)
cv_scores_rus = cross_val_score(lr_model_rus, X_train_rus, y_train_
print("\n[Pipeline 3: Random Undersampling]")
print("RUS-balanced X_train_logistic shape:", X_train_rus.shape)
print("CV Scores:", cv_scores_rus)
print("Average recall:", cv_scores_rus.mean())
```

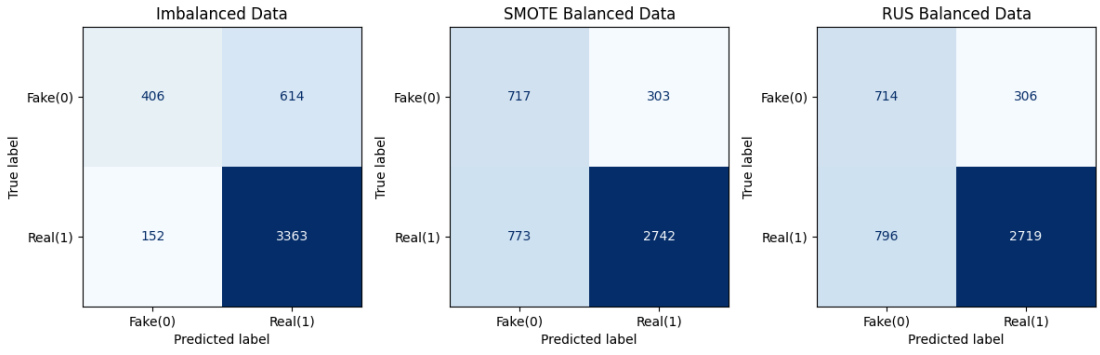
```
[Pipeline 1: Imbalanced]
Unbalanced X_train_logistic shape: (10581, 100)
CV Scores: [0.69278891 0.66515936 0.69772238 0.67549447 0.6918291
2]
Average recall: 0.6845988454089068

[Pipeline 2: SMOTE Oversampling]
SMOTE-balanced X_train_logistic shape: (16404, 100)
CV Scores: [0.75739808 0.75648344 0.76104342 0.77110291 0.7704268
3]
Average recall: 0.7632909365199685

[Pipeline 3: Random Undersampling]
RUS-balanced X_train_logistic shape: (4758, 100)
CV Scores: [0.75       0.75105042 0.75       0.7434498  0.7455019
9]
Average recall: 0.7480004422821761
```

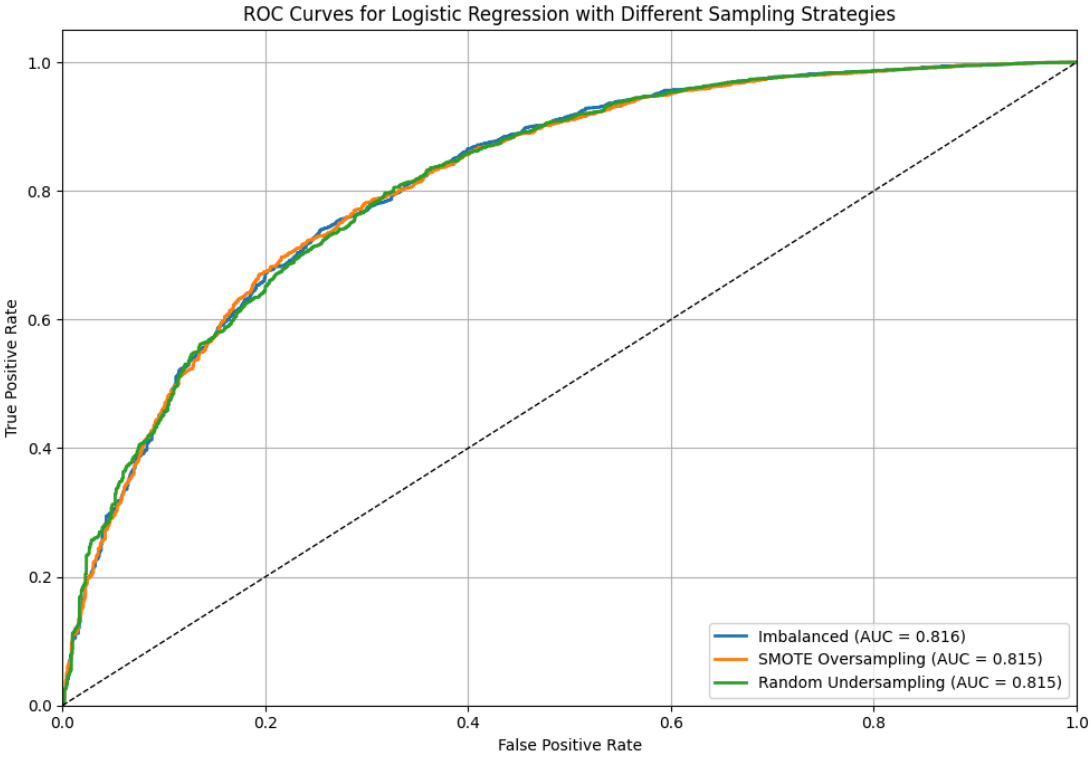
In [118]:

```
# Predictions by model fitted with imbalanced data
lr_model_imbalanced.fit(X_train_logistic, y_train_logistic)
y_pred_imbalanced = lr_model_imbalanced.predict(X_test_1)
print("\nClassification Report : Logistic Regression fitted on imba
print(classification_report(y_test, y_pred_imbalanced, target_names
y_proba_imbalanced = lr_model_imbalanced.predict_proba(X_test_1)[:,
fpr_imbalanced, tpr_imbalanced, _ = roc_curve(y_test, y_proba_imbal
roc_auc_imbalanced = auc(fpr_imbalanced, tpr_imbalanced)

# Predictions by model fitted with SMOTE balanced data
lr_model_smote.fit(X_train_smote, y_train_smote)
y_pred_smote_balanced = lr_model_smote.predict(X_test_1)
print("\nClassification Report : Logistic Regression fitted on SMOT
print(classification_report(y_test, y_pred_smote_balanced, target_r
y_proba_smote = lr_model_smote.predict_proba(X_test_1)[:, 1]
fpr_smote, tpr_smote, _ = roc_curve(y_test, y_proba_smote)
roc_auc_smote = auc(fpr_smote, tpr_smote)

# Predictions by model fitted with RUS balanced data
lr_model_rus.fit(X_train_rus, y_train_rus)
y_pred_rus_balanced = lr_model_rus.predict(X_test_1)  # Fixed bug h
print("\nClassification Report : Logistic Regression fitted on RUS
print(classification_report(y_test, y_pred_rus_balanced, target_nam
y_proba_rus = lr_model_rus.predict_proba(X_test_1)[:, 1]
fpr_rus, tpr_rus, _ = roc_curve(y_test, y_proba_rus)
roc_auc_rus = auc(fpr_rus, tpr_rus)

# Plot Confusion Matrices
fig, axes = plt.subplots(1, 3, figsize=(12, 5))
cm1 = confusion_matrix(y_test, y_pred_imbalanced)
disp1 = ConfusionMatrixDisplay(confusion_matrix=cm1, display_labels
disp1.plot(ax=axes[0], cmap='Blues', colorbar=False)
axes[0].set_title("Imbalanced Data")
cm2 = confusion_matrix(y_test, y_pred_smote_balanced)
disp2 = ConfusionMatrixDisplay(confusion_matrix=cm2, display_labels
disp2.plot(ax=axes[1], cmap='Blues', colorbar=False)
axes[1].set_title("SMOTE Balanced Data")
cm3 = confusion_matrix(y_test, y_pred_rus_balanced)
disp3 = ConfusionMatrixDisplay(confusion_matrix=cm3, display_labels
disp3.plot(ax=axes[2], cmap='Blues', colorbar=False)
axes[2].set_title("RUS Balanced Data")
plt.tight_layout()
plt.savefig('/kaggle/working/images/logistic-regression-confusion-m
plt.show()

# Plot ROC-AUC curves
plt.figure(figsize=(10, 7))
plt.plot(fpr_imbalanced, tpr_imbalanced, label=f"Imbalanced (AUC =
plt.plot(fpr_smote, tpr_smote, label=f"SMOTE Oversampling (AUC = {r
plt.plot(fpr_rus, tpr_rus, label=f"Random Undersampling (AUC = {roc
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curves for Logistic Regression with Different Sampli
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.savefig('/kaggle/working/images/logistic-regression-ROC-AUC.png
```

```
plt.show()
```

```
Classification Report : Logistic Regression fitted on imbalanced
data
              precision    recall  f1-score   support

     Fake(0)     0.7276    0.3980    0.5146      1020
    Real (1)     0.8456    0.9568    0.8978      3515

    accuracy                         0.8311      4535
   macro avg     0.7866    0.6774    0.7062      4535
weighted avg     0.8191    0.8311    0.8116      4535


Classification Report : Logistic Regression fitted on SMOTE balan
ced data
              precision    recall  f1-score   support

     Fake(0)     0.4812    0.7029    0.5713      1020
    Real (1)     0.9005    0.7801    0.8360      3515

    accuracy                         0.7627      4535
   macro avg     0.6909    0.7415    0.7036      4535
weighted avg     0.8062    0.7627    0.7764      4535


Classification Report : Logistic Regression fitted on RUS balance
d data
              precision    recall  f1-score   support

     Fake(0)     0.4728    0.7000    0.5644      1020
    Real (1)     0.8988    0.7735    0.8315      3515

    accuracy                         0.7570      4535
   macro avg     0.6858    0.7368    0.6980      4535
weighted avg     0.8030    0.7570    0.7714      4535
```





The baseline model fitted with SMOTE-balanced data yields better performance since the aim is to catch as many fake news articles as possible.

- Its ***recall*** score for the minority class `Fake(0)` is highest at 0.7029.
- Its ***macro-averaged recall*** is highest at 0.7415.

Objective 3: Build an XGBoost model, LSTM neural network, and Finetune the RoBERTa-base Transformer

XGBoost Model

In [119]:

```
# Prepare training and validation sets
X_train_xgb = np.vstack((X_train_1, X_val_1))
y_train_xgb = np.hstack((y_train, y_val))

# Define pipeline for fitting model with imbalanced data
xgb_model_imbalanced = XGBClassifier(use_label_encoder=False, eval_
cv_scores_imbalanced = cross_val_score(xgb_model_imbalanced, X_trai
print("\n[Pipeline 1: Imbalanced]")
print("Unbalanced X_train_xgb shape:", X_train_xgb.shape)
print("CV Scores:", cv_scores_imbalanced)
print("Average recall:", cv_scores_imbalanced.mean())

# Define pipeline for fitting model with oversampled minority class
smote = SMOTE(sampling_strategy='minority', random_state=42)
X_train_smote, y_train_smote = smote.fit_resample(X_train_xgb, y_tr
xgb_model_smote = XGBClassifier(use_label_encoder=False, eval_metri
cv_scores_smote = cross_val_score(xgb_model_smote, X_train_smote, y
print("\n[Pipeline 2: SMOTE Oversampling]")
print("SMOTE-balanced X_train_xgb shape:", X_train_smote.shape)
print("CV Scores:", cv_scores_smote)
print("Average recall:", cv_scores_smote.mean())

# Define pipeline for fitting model with undersampled majority clas
rus = RandomUnderSampler(sampling_strategy='majority', random_state
X_train_rus, y_train_rus = rus.fit_resample(X_train_xgb, y_train_xg
xgb_model_rus = XGBClassifier(use_label_encoder=False, eval_metric=
cv_scores_rus = cross_val_score(xgb_model_rus, X_train_rus, y_train
print("\n[Pipeline 3: Random Undersampling]")
print("RUS-balanced X_train_xgb shape:", X_train_rus.shape)
print("CV Scores:", cv_scores_rus)
print("Average recall:", cv_scores_rus.mean())
```

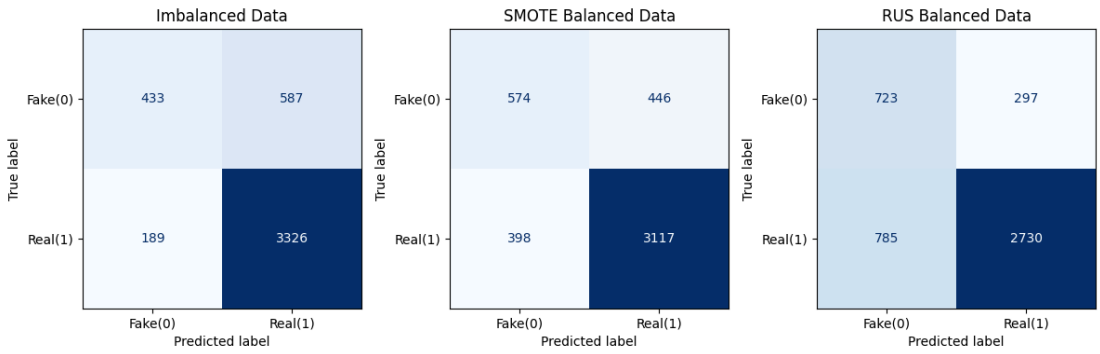
```
[Pipeline 1: Imbalanced]
Unbalanced X_train_xgb shape: (10581, 100)
CV Scores: [0.69994149 0.6917273  0.71131123 0.70792171 0.7225222
1]
Average recall: 0.7066847884361707

[Pipeline 2: SMOTE Oversampling]
SMOTE-balanced X_train_xgb shape: (16404, 100)
CV Scores: [0.86163739 0.87838822 0.91406341 0.92564506 0.9189024
4]
Average recall: 0.8997273004265691

[Pipeline 3: Random Undersampling]
RUS-balanced X_train_xgb shape: (4758, 100)
CV Scores: [0.75210084 0.75420168 0.75210084 0.74029854 0.7297479
]
Average recall: 0.7456899601946041
```

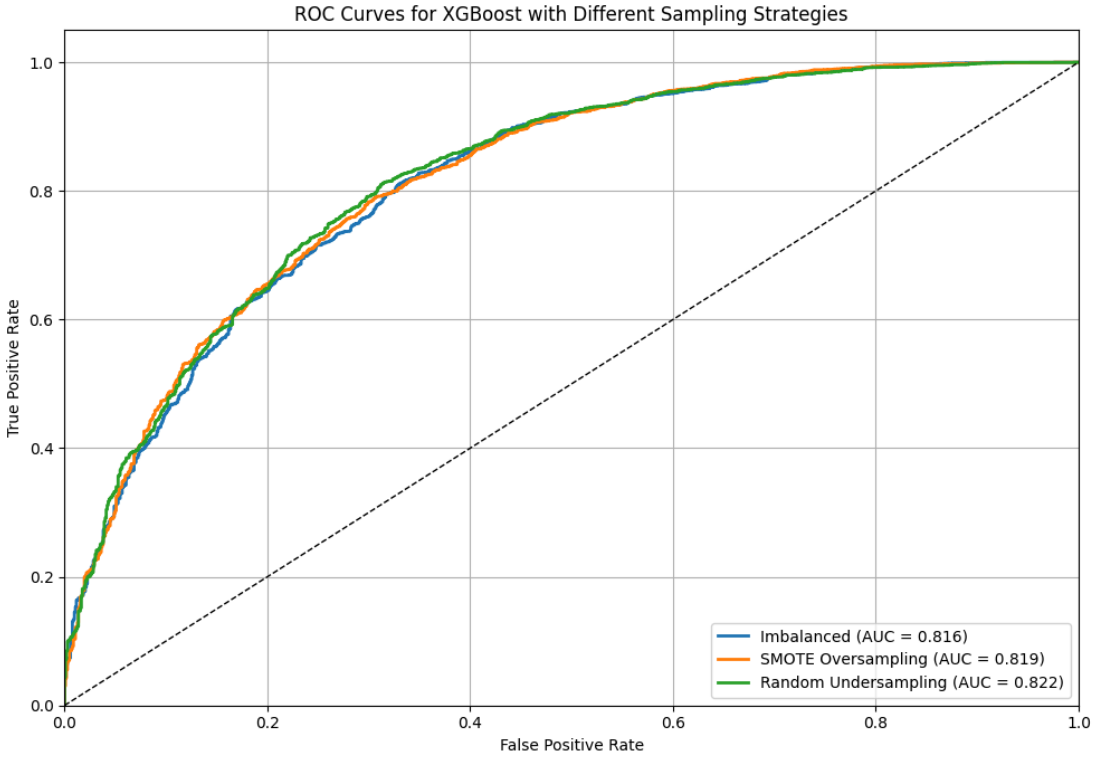
In [120]:

```
# Predictions by model fitted with imbalanced data
xgb_model_imbalanced.fit(X_train_xgb, y_train_xgb)
y_pred_imbalanced = xgb_model_imbalanced.predict(X_test_1)
print("\nClassification Report : XGBoost fitted on imbalanced data"
print(classification_report(y_test, y_pred_imbalanced, target_names
y_proba_imbalanced = xgb_model_imbalanced.predict_proba(X_test_1)[
fpr_imbalanced, tpr_imbalanced, _ = roc_curve(y_test, y_proba_imbal
roc_auc_imbalanced = auc(fpr_imbalanced, tpr_imbalanced)

# Predictions by model fitted with SMOTE balanced data
xgb_model_smote.fit(X_train_smote, y_train_smote)
y_pred_smote_balanced = xgb_model_smote.predict(X_test_1)
print("\nClassification Report : XGBoost fitted on SMOTE balanced d
print(classification_report(y_test, y_pred_smote_balanced, target_r
y_proba_smote = xgb_model_smote.predict_proba(X_test_1)[:, 1]
fpr_smote, tpr_smote, _ = roc_curve(y_test, y_proba_smote)
roc_auc_smote = auc(fpr_smote, tpr_smote)

# Predictions by model fitted with RUS balanced data
xgb_model_rus.fit(X_train_rus, y_train_rus)
y_pred_rus_balanced = xgb_model_rus.predict(X_test_1)
print("\nClassification Report : XGBoost fitted on RUS balanced dat
print(classification_report(y_test, y_pred_rus_balanced, target_nam
y_proba_rus = xgb_model_rus.predict_proba(X_test_1)[:, 1]
fpr_rus, tpr_rus, _ = roc_curve(y_test, y_proba_rus)
roc_auc_rus = auc(fpr_rus, tpr_rus)

# Plot Confusion Matrices
fig, axes = plt.subplots(1, 3, figsize=(12, 5))
cm1 = confusion_matrix(y_test, y_pred_imbalanced)
disp1 = ConfusionMatrixDisplay(confusion_matrix=cm1, display_labels
disp1.plot(ax=axes[0], cmap='Blues', colorbar=False)
axes[0].set_title("Imbalanced Data")
cm2 = confusion_matrix(y_test, y_pred_smote_balanced)
disp2 = ConfusionMatrixDisplay(confusion_matrix=cm2, display_labels
disp2.plot(ax=axes[1], cmap='Blues', colorbar=False)
axes[1].set_title("SMOTE Balanced Data")
cm3 = confusion_matrix(y_test, y_pred_rus_balanced)
disp3 = ConfusionMatrixDisplay(confusion_matrix=cm3, display_labels
disp3.plot(ax=axes[2], cmap='Blues', colorbar=False)
axes[2].set_title("RUS Balanced Data")
plt.tight_layout()
plt.savefig('/kaggle/working/images/xgboost-confusion-matrices.png
plt.show()

# Plot ROC-AUC curves
plt.figure(figsize=(10, 7))
plt.plot(fpr_imbalanced, tpr_imbalanced, label=f"Imbalanced (AUC =
plt.plot(fpr_smote, tpr_smote, label=f"SMOTE Oversampling (AUC = {r
plt.plot(fpr_rus, tpr_rus, label=f"Random Undersampling (AUC = {roc
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curves for XGBoost with Different Sampling Strategie
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.savefig('/kaggle/working/images/xgboost-ROC-AUC.png', dpi=300,
```

```
plt.show()
```

```
Classification Report : XGBoost fitted on imbalanced data
              precision    recall  f1-score   support

     Fake(0)     0.6961    0.4245    0.5274      1020
    Real (1)     0.8500    0.9462    0.8955      3515

    accuracy                         0.8289      4535
   macro avg     0.7731    0.6854    0.7115      4535
weighted avg     0.8154    0.8289    0.8127      4535


Classification Report : XGBoost fitted on SMOTE balanced data
              precision    recall  f1-score   support

     Fake(0)     0.5905    0.5627    0.5763      1020
    Real (1)     0.8748    0.8868    0.8808      3515

    accuracy                         0.8139      4535
   macro avg     0.7327    0.7248    0.7285      4535
weighted avg     0.8109    0.8139    0.8123      4535


Classification Report : XGBoost fitted on RUS balanced data
              precision    recall  f1-score   support

     Fake(0)     0.4794    0.7088    0.5720      1020
    Real (1)     0.9019    0.7767    0.8346      3515

    accuracy                         0.7614      4535
   macro avg     0.6907    0.7427    0.7033      4535
weighted avg     0.8069    0.7614    0.7755      4535
```





The XGBoost model fitted with RUS-balanced data yields better performance since the aim is to catch as many fake news articles as possible.

- Its ***recall*** score for the minority class `Fake(0)` is highest at 0.7088.
- Its ***macro-averaged recall*** is highest at 0.7427.

LSTM Neural Network

In [140]:

```
# Define Parameters
max_len = 300  # Max tokens per sample
embedding_dim = 100  # As set in Word2Vec

# Define function to convert the text feature to vector sequences
def text_to_vector_sequence(text, model, max_len=300, embedding_dim
    tokens = word_tokenize(text.lower())
    sequence = [model.wv[token] for token in tokens if token in mod
    if len(sequence) == 0:
        return np.zeros((max_len, embedding_dim))
    sequence = sequence[:max_len]  # Truncate
    if len(sequence) < max_len:  # Pad
        padding = np.zeros((max_len - len(sequence), embedding_dim)
        sequence = np.vstack((sequence, padding))
    return np.array(sequence)

# Define function to compute f1-score metric
class F1Score(tf.keras.metrics.Metric):
    def __init__(self, name='f1_score', **kwargs):
        super().__init__(name=name, **kwargs)
        self.precision = Precision()
        self.recall = Recall()
    def update_state(self, y_true, y_pred, sample_weight=None):
        y_pred_binary = tf.round(y_pred)
        self.precision.update_state(y_true, y_pred_binary, sample_w
        self.recall.update_state(y_true, y_pred_binary, sample_weig
    def result(self):
        p = self.precision.result()
        r = self.recall.result()
        return 2 * ((p * r) / (p + r + tf.keras.backend.epsilon()))
    def reset_state(self):
        self.precision.reset_state()
        self.recall.reset_state()

# Define function build the LSTM models' architecture
def build_lstm_model(max_len, embedding_dim):
    inputs = Input(shape=(max_len, embedding_dim))
    x = Masking(mask_value=0.0)(inputs)
    x = LSTM(64)(x)
    x = Dropout(0.3)(x)
    x = Dense(32, activation='relu')(x)
    outputs = Dense(1, activation='sigmoid')(x)
    model = Model(inputs, outputs)
    return model
```

In [141]:

```
# Define function to train and evaluate the models' respective perf
def train_and_evaluate_pipeline(
    X_train_text, X_val_text, X_test_text,
    y_train, y_val, y_test,
    word2vec_model, max_len, embedding_dim,
    sampling_strategy=None, name="Imbalanced"):
    print(f"\n Training on {name} dataset")

    # Convert the text feature to sequences
    X_train_seq = np.array([text_to_vector_sequence(text, word2vec_
    X_val_seq = np.array([text_to_vector_sequence(text, word2vec_mo
    X_test_seq = np.array([text_to_vector_sequence(text, word2vec_m

    # Reshape for resampling (flatten to 2D)
    if sampling_strategy:
        X_train_flat = X_train_seq.reshape((X_train_seq.shape[0], -
        X_val_flat = X_val_seq.reshape((X_val_seq.shape[0], -1))
        if sampling_strategy == "smote":
            sampler = SMOTE(sampling_strategy='minority', random_st
        elif sampling_strategy == "rus":
            sampler = RandomUnderSampler(sampling_strategy='majorit
        else:
            raise ValueError("sampling_strategy must be 'smote' or
        X_train_resampled, y_train_resampled = sampler.fit_resample
        X_val_resampled, y_val_resampled = sampler.fit_resample(X_v
        # Reshape back to 3D
        X_train_seq = X_train_resampled.reshape((-1, max_len, embed
        X_val_seq = X_val_resampled.reshape((-1, max_len, embedding
        y_train_used, y_val_used = y_train_resampled, y_val_resampl
    else:
        y_train_used, y_val_used = y_train, y_val

    # Build and compile model
    model = build_lstm_model(max_len, embedding_dim)
    model.compile(
        optimizer='adam',
        loss='binary_crossentropy',
        metrics=[
            Precision(name='precision'),
            Recall(name='recall'),
            F1Score(name='f1_score')])

    # Train
    history = model.fit(
        X_train_seq, y_train_used,
        validation_data=(X_val_seq, y_val_used),
        epochs=6,
        batch_size=64,
        verbose=1)

    # Extract metrics per epoch
    metrics_df = pd.DataFrame({
        'epoch': range(1, len(history.history['loss']) + 1),
        'train_loss': history.history['loss'],
        'val_loss': history.history['val_loss'],
        'val_precision': history.history['val_precision'],
        'val_recall': history.history['val_recall'],
        'val_f1_score': history.history['f1_score']})
    print(f"\n Metrics per epoch - {name}:")
    print(metrics_df.round(4))
```

```
    # Plot Train vs Val Loss plots
    plt.figure(figsize=(8, 5))
    plt.plot(metrics_df['epoch'], metrics_df['train_loss'], label=
    plt.plot(metrics_df['epoch'], metrics_df['val_loss'], label='Va
    plt.title(f'Training vs Validation Loss - {name}')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.tight_layout()

    # Save training loss vs. validation loss plots to /kaggle/worki
    save_name = name.replace(" ", "_").replace("-", "_")
    plot_path = os.path.join(images_dir, f"LSTM-loss-curve-({save_n
    plt.savefig(plot_path)
    plt.show()

    # Test set prediction
    y_test_pred_proba = model.predict(X_test_seq).ravel()
    y_test_pred = (y_test_pred_proba > 0.5).astype(int)
    return history, metrics_df, model, y_test_pred_proba, y_test_pr
```

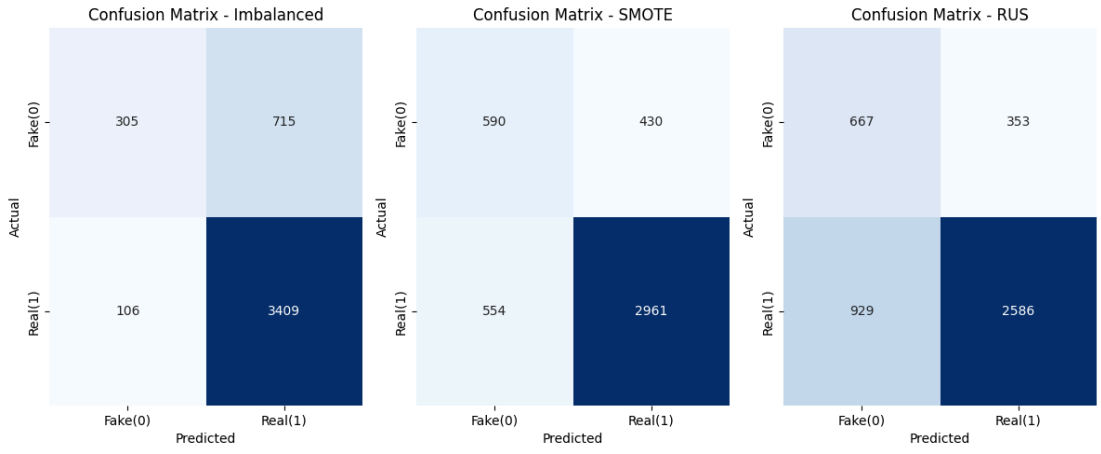
In [142]:

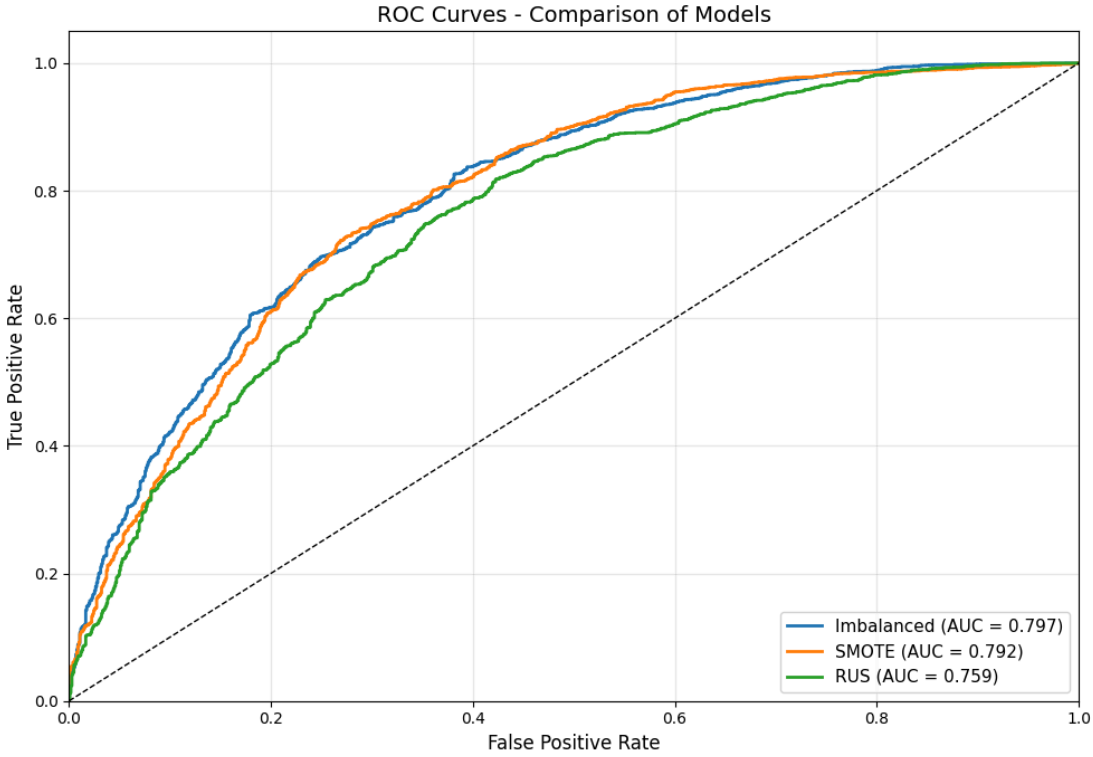
```
# Run pipelines
results = {}

# Imbalanced
print("="*60)
history_imb, df_imb, model_imb, y_test_proba_imb, y_test_pred_imb,
    X_train['text'], X_val['text'], X_test['text'],
    y_train, y_val, y_test,
    word2vec_model, max_len, embedding_dim,
    sampling_strategy=None,
    name="Imbalanced")
results['Imbalanced'] = (y_test_true, y_test_pred_imb, y_test_proba

# SMOTE
print("="*60)
history_smote, df_smote, model_smote, y_test_proba_smote, y_test_pr
    X_train['text'], X_val['text'], X_test['text'],
    y_train, y_val, y_test,
    word2vec_model, max_len, embedding_dim,
    sampling_strategy="smote",
    name="SMOTE Balanced")
results['SMOTE'] = (y_test_true, y_test_pred_smote, y_test_proba_sm

# RUS
print("="*60)
history_rus, df_rus, model_rus, y_test_proba_rus, y_test_pred_rus,
    X_train['text'], X_val['text'], X_test['text'],
    y_train, y_val, y_test,
    word2vec_model, max_len, embedding_dim,
    sampling_strategy="rus",
    name="RUS Balanced")
results['RUS'] = (y_test_true, y_test_pred_rus, y_test_proba_rus)
```

```
============================================================

 Training on Imbalanced dataset
Epoch 1/6
116/116 ━━━━━━━━━━━━━━━━━━━━ 8s 38ms/step - f1_score: 0.8754 - l
oss: 0.5238 - precision: 0.7849 - recall: 0.9897 - val_f1_score:
0.8852 - val_loss: 0.4785 - val_precision: 0.8014 - val_recall:
0.9886
Epoch 2/6
116/116 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - f1_score: 0.8855 - l
oss: 0.4593 - precision: 0.8070 - recall: 0.9810 - val_f1_score:
0.8801 - val_loss: 0.4558 - val_precision: 0.8231 - val_recall:
0.9456
Epoch 3/6
116/116 ━━━━━━━━━━━━━━━━━━━━ 2s 20ms/step - f1_score: 0.8928 - l
oss: 0.4182 - precision: 0.8312 - recall: 0.9644 - val_f1_score:
0.8881 - val_loss: 0.4397 - val_precision: 0.8196 - val_recall:
0.9691
Epoch 4/6
```

In [143]:

```
# Plot confusion matrices
fig, axes = plt.subplots(1, 3, figsize=(12, 5))
class_names = ['Fake(0)', 'Real(1)']
datasets = ['Imbalanced-LSTM', 'SMOTE-LSTM', 'RUS-LSTM']
for ax, (name, (y_true, y_pred, _)) in zip(axes, results.items()):
    cm = confusion_matrix(y_true, y_pred)
    sns.heatmap(
        cm,
        annot=True,
        fmt='d',
        cmap='Blues',
        ax=ax,
        cbar=False,
        xticklabels=class_names,
        yticklabels=class_names)
    ax.set_title(f'Confusion Matrix - {name}', fontsize=12)
    ax.set_xlabel('Predicted', fontsize=10)
    ax.set_ylabel('Actual', fontsize=10)
plt.tight_layout()
plt.savefig('/kaggle/working/images/LSTM-confusion_matrices.png', c
plt.show()

# Plot ROC-AUC curves
plt.figure(figsize=(10, 7))
for name, (y_true, _, y_proba) in results.items():
    fpr, tpr, _ = roc_curve(y_true, y_proba)
    roc_auc = auc(fpr, tpr)
    plt.plot(fpr, tpr, lw=2, label=f'{name} (AUC = {roc_auc:.3f})')
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate', fontsize=12)
plt.ylabel('True Positive Rate', fontsize=12)
plt.title('ROC Curves - Comparison of Models', fontsize=14)
plt.legend(loc="lower right", fontsize=11)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('/kaggle/working/images/LSTM-ROC-AUC.png', dpi=300, bbo
plt.show()
```




ROC Curves - Comparison of Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Imbalanced (AUC = 0.797)
SMOTE (AUC = 0.792)
RUS (AUC = 0.759)

In [144]:
```
# Print per-class and macro/weighted averages
for name, (y_true, y_pred, y_proba) in results.items():
    print(f"\n {name} Model:")
    print("-" * 40)
    print(classification_report(y_true, y_pred, target_names=['Fake
```

```
 Imbalanced Model:
----------------------------------------
              precision    recall  f1-score   support

     Fake(0)     0.7421    0.2990    0.4263      1020
     Real(1)     0.8266    0.9698    0.8925      3515

    accuracy                         0.8190      4535
   macro avg     0.7844    0.6344    0.6594      4535
weighted avg     0.8076    0.8190    0.7877      4535


 SMOTE Model:
----------------------------------------
              precision    recall  f1-score   support

     Fake(0)     0.5157    0.5784    0.5453      1020
     Real(1)     0.8732    0.8424    0.8575      3515

    accuracy                         0.7830      4535
   macro avg     0.6945    0.7104    0.7014      4535
weighted avg     0.7928    0.7830    0.7873      4535


 RUS Model:
----------------------------------------
              precision    recall  f1-score   support

     Fake(0)     0.4179    0.6539    0.5099      1020
     Real(1)     0.8799    0.7357    0.8014      3515

    accuracy                         0.7173      4535
   macro avg     0.6489    0.6948    0.6557      4535
weighted avg     0.7760    0.7173    0.7358      4535
```

The LSTM model fitted with SMOTE-balanced data yields better performance since the aim is to catch as many fake news articles as possible. Its ***macro-averaged recall*** is highest at 0.7104.

Finetune the RoBERTa-base Transformer

In [130]:

```
# Set up device, pretrained transformer, and tokenizer

device = torch.device("cuda" if torch.cuda.is_available() else "cpu
print(f"Using device: {device}")

model_name = "roberta-base"  # Set up pretrained transformer
roberta_tokenizer = RobertaTokenizer.from_pretrained(model_name) #
```

```
Using device: cuda
```

In [131]:

```
# Initialize the NewsDataset with text chunking using a sliding wi
class NewsDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_length=512, st
        self.tokenizer = tokenizer
        self.max_length = max_length
        self.stride = stride
        self.all_chunk_encodings = []      # Store {'input_ids', 'a
        self.all_chunk_labels = []         # Store the label for ea
        self.chunk_original_indices = []   # Store the index of the

        # Process each original text to create chunks
        for original_idx, text in enumerate(texts):
            tokenized_output = tokenizer(
                text,
                truncation=False,
                padding=False,
                max_length=self.max_length,
                stride=self.stride,
                return_overflowing_tokens=True,
                return_tensors="pt")      # Return PyTorch tensors

            # Iterate through each chunk generated for the current
            for i in range(len(tokenized_output['input_ids'])):
                chunk_ids = tokenized_output['input_ids'][i]
                chunk_mask = tokenized_output['attention_mask'][i]

                if chunk_ids.shape[0] > max_length:        # Trunca
                    chunk_ids = chunk_ids[:max_length]
                    chunk_mask = chunk_mask[:max_length]
                elif chunk_ids.shape[0] < max_length:      # Pad i
                    pad_len = max_length - chunk_ids.shape[0]
                    pad_tensor = torch.full((pad_len,), tokenizer.p
                    mask_pad = torch.zeros(pad_len, dtype=torch.lon
                    chunk_ids = torch.cat([chunk_ids, pad_tensor])
                    chunk_mask = torch.cat([chunk_mask, mask_pad])

                self.all_chunk_encodings.append({
                    'input_ids': chunk_ids,
                    'attention_mask': chunk_mask})
                self.all_chunk_labels.append(labels[original_idx])
                self.chunk_original_indices.append(original_idx)

    # Return the total number of chunks across all original articl
    def __len__(self):
        return len(self.all_chunk_labels)

    # Retrieve a single chunk's data and its associated original la
    def __getitem__(self, idx):
        item = {key: val.clone() for key, val in self.all_chunk_enc
        item['labels'] = torch.tensor(self.all_chunk_labels[idx], d
        item['original_index'] = torch.tensor(self.chunk_original_i
        return item
```

In [132]:

```
# Define function to resample imbalanced data before chunking

def resample_data(X_texts, y_labels, strategy="smote"):
    X_list = X_texts.tolist()
    y_array = np.array(y_labels)

    if strategy == "smote": # Oversample the minority class

        # Use dummy features (lengths) to simulate feature space
        X_dummy = np.array([[len(text)] for text in X_list])
        sampler = SMOTE(sampling_strategy='minority', random_state=
        _, y_resampled = sampler.fit_resample(X_dummy, y_array)

        # Resample texts by oversampling the minority class
        from collections import Counter
        counter = Counter(y_array)
        target_count = max(counter.values())
        X_resampled, y_resampled = [], []
        for label in [0, 1]:
            indices = np.where(y_array == label)[0]
            resampled_indices = np.random.choice(indices, size=targ
            X_resampled.extend([X_list[i] for i in resampled_indice
            y_resampled.extend([y_array[i] for i in resampled_indic

    elif strategy == "rus": # Undersample the majority class
        X_dummy = np.array([[len(text)] for text in X_list])
        sampler = RandomUnderSampler(sampling_strategy='majority',
        _, y_resampled = sampler.fit_resample(X_dummy, y_array)

        # Resample texts by undersampling the majority cl according
        from collections import Counter
        counter = Counter(y_array)
        target_count = min(counter.values())
        X_resampled, y_resampled = [], []
        for label in [0, 1]:
            indices = np.where(y_array == label)[0]
            resampled_indices = np.random.choice(indices, size=targ
            X_resampled.extend([X_list[i] for i in resampled_indice
            y_resampled.extend([y_array[i] for i in resampled_indic

    else:  # imbalanced dataset
        X_resampled, y_resampled = X_list, y_array

    return X_resampled, y_resampled
```

In [133]:

```
# Initialize the MetricsTracker class
class MetricsTracker:
    def __init__(self):
        self.epoch_data = []
    def log_epoch(self, epoch, metrics):
        self.epoch_data.append({'epoch': int(epoch), **metrics})
    def get_dataframe(self):
        return pd.DataFrame(self.epoch_data).set_index('epoch')
metrics_tracker = MetricsTracker()

# Define function to compute performance metrics
def compute_metrics(p):
    preds = np.argmax(p.predictions, axis=1)
    probs = torch.softmax(torch.tensor(p.predictions), dim=1).numpy
    return {
        'accuracy': accuracy_score(p.label_ids, preds),
        'precision': precision_score(p.label_ids, preds, zero_divis
        'recall': recall_score(p.label_ids, preds, zero_division=0)
        'f1': f1_score(p.label_ids, preds, zero_division=0),
        'roc_auc': roc_auc_score(p.label_ids, probs)}

# Initialize the CustomTrainer class for logging
class CustomTrainer(Trainer):
    def evaluate(self, *args, **kwargs):
        metrics = super().evaluate(*args, **kwargs)
        self.control = self.callback_handler.on_evaluate(self.args,
        metrics_tracker.log_epoch(self.state.epoch, metrics)
        return metrics

# Define function to extract average training loss per epoch from t
def extract_per_epoch_train_loss(trainer):
    train_loss_epochs = {}
    for log in trainer.state.log_history:
        if 'loss' in log and 'epoch' in log:
            epoch = round(log['epoch'])  # Round float epoch to nea
            if epoch not in train_loss_epochs:
                train_loss_epochs[epoch] = []
            train_loss_epochs[epoch].append(log['loss'])
    # Return average loss per epoch
    return pd.Series({epoch: np.mean(losses) for epoch, losses in t

# Define training arguments function
def get_training_args(run_name):
    return TrainingArguments(
        output_dir=f"./roberta_results_{run_name}",
        run_name=f"roberta-{run_name}",
        num_train_epochs=4,
        per_device_train_batch_size=16,
        per_device_eval_batch_size=32,
        learning_rate=2e-5,
        warmup_ratio=0.06,
        weight_decay=0.01,
        eval_strategy="epoch",
        save_strategy="epoch",
        load_best_model_at_end=True,
        metric_for_best_model="f1",
        greater_is_better=True,
        logging_dir=f"./roberta_logs_{run_name}",
        logging_steps=100,
        seed=42,
        fp16=torch.cuda.is_available(),
```

```
        report_to="none",
        save_total_limit=2)
```

In [134]:

```
# Define function to implement finetuning pipelines

def finetune_roberta_pipeline(X_train, y_train, X_val, y_val, X_tes
    print(f"\n Fine-tuning RoBERTa ({name})")
    X_train_res, y_train_res = resample_data(X_train, y_train, stra

    # Create chunked datasets
    train_dataset = NewsDataset(X_train_res, y_train_res, roberta_t
    val_dataset = NewsDataset(X_val.tolist(), y_val.tolist(), robe
    test_dataset = NewsDataset(X_test.tolist(), y_test.tolist(), ro

    # Load model
    model = RobertaForSequenceClassification.from_pretrained(
        model_name,
        num_labels=2,
        hidden_dropout_prob=0.2,
        attention_probs_dropout_prob=0.2
    ).to(device)

    # Reset tracker and trainer
    global metrics_tracker
    metrics_tracker = MetricsTracker()
    trainer_args = get_training_args(name.replace(" ", "_"))
    trainer = CustomTrainer(
        model=model,
        args=trainer_args,
        train_dataset=train_dataset,
        eval_dataset=val_dataset,
        compute_metrics=compute_metrics,)

    # Train and compile the performance metrics DataFrame
    trainer.train()
    avg_train_loss = extract_per_epoch_train_loss(trainer)
    metrics_df = metrics_tracker.get_dataframe()
    metrics_df = metrics_df.assign(
        train_loss=metrics_df.index.map(avg_train_loss),  # Map and
        val_loss=metrics_df['eval_loss'])
    print(f"\nMetrics per epoch - {name}:")
    print(metrics_df[['train_loss', 'val_loss', 'eval_precision',

    # Plot and Save Loss Curve
    plt.figure(figsize=(8, 5))
    epochs = metrics_df.index
    plt.plot(epochs, metrics_df['train_loss'], label='Train Loss',
    plt.plot(epochs, metrics_df['val_loss'], label='Validation Loss
    plt.title(f'Training vs Validation Loss - RoBERTa ({name})')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plot_path = f"/kaggle/working/images/roberta_loss_{name.replace
    plt.savefig(plot_path)
    plt.show()

    # Evaluate respective performance on the test set
    test_outs = trainer.predict(test_dataset)
    y_test_pred = np.argmax(test_outs.predictions, axis=1)
    y_test_proba = torch.softmax(torch.tensor(test_outs.predictions

    return metrics_df, model, y_test_pred, y_test_proba, y_test
```

In [135]:

```
# Run Pipelines
results = {}

# Imbalanced
print("="*60)
df_imb, model_imb, y_test_pred_imb, y_test_proba_imb, y_test_true =
    X_train['text'], y_train, X_val['text'], y_val, X_test['text'],
    strategy=None, name="Imbalanced")
results['RoBERTa-Imbalanced'] = (y_test_true, y_test_pred_imb, y_te

# SMOTE
print("="*60)
df_smote, model_smote, y_test_pred_smote, y_test_proba_smote, _ = f
    X_train['text'], y_train, X_val['text'], y_val, X_test['text'],
    strategy="smote", name="SMOTE")
results['RoBERTa-SMOTE'] = (y_test_true, y_test_pred_smote, y_test_

# RUS
print("="*60)
df_rus, model_rus, y_test_pred_rus, y_test_proba_rus, _ = finetune_
    X_train['text'], y_train, X_val['text'], y_val, X_test['text'],
    strategy="rus", name="RUS")
results['RoBERTa-RUS'] = (y_test_true, y_test_pred_rus, y_test_prob
```

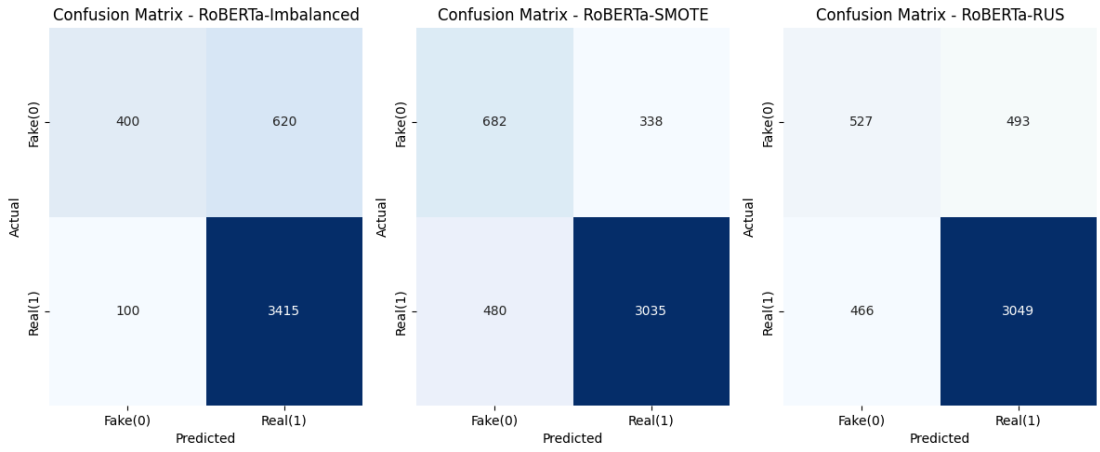
```
============================================================

 Fine-tuning RoBERTa (Imbalanced)

Some weights of RobertaForSequenceClassification were not initia
lized from the model checkpoint at roberta-base and are newly in
itialized: ['classifier.dense.bias', 'classifier.dense.weight',
'classifier.out_proj.bias', 'classifier.out_proj.weight']
You should probably TRAIN this model on a down-stream task to be
able to use it for predictions and inference.
```

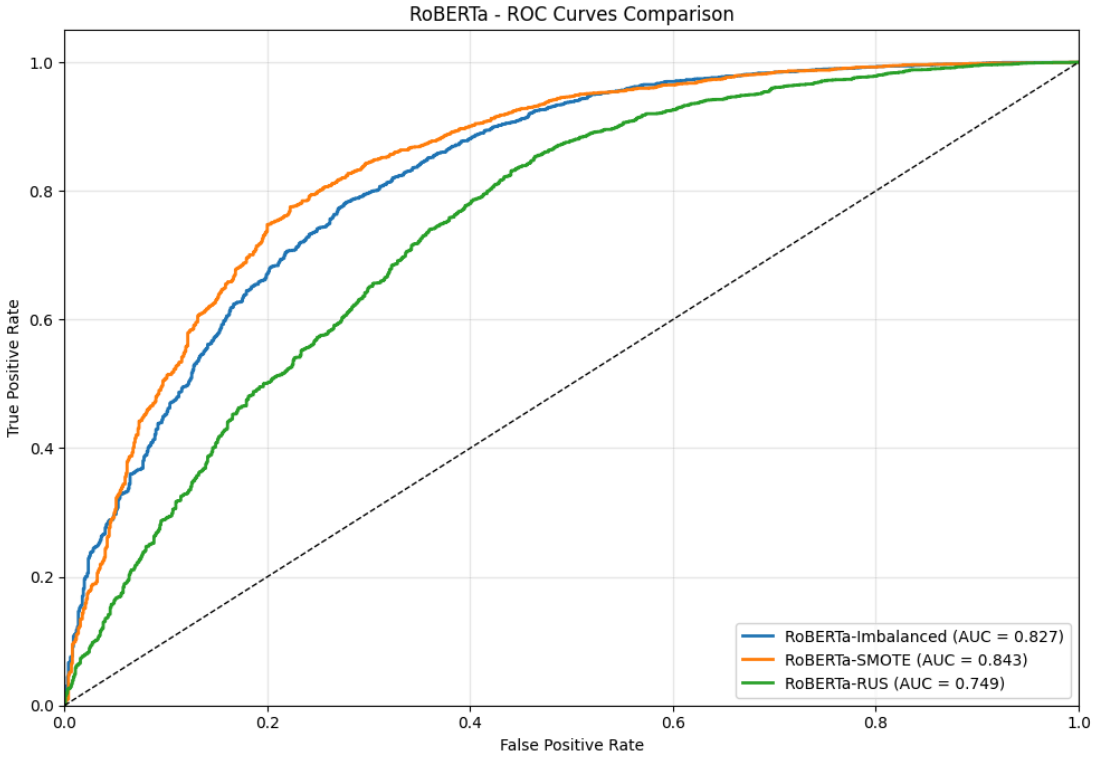
[928/928 31:32, Epoch 4/4]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1	Roc Auc
1	0.488200	0.414061	0.824567	0.867284	0.913450	0.889768	0.813613
2	0.404200	0.404004	0.848819	0.849876	0.977651	0.909297	0.839643
3	0.350900	0.412760	0.833071	0.892320	0.892320	0.892320	0.846552

In [136]:

```
# Plot Confusion Matrices
fig, axes = plt.subplots(1, 3, figsize=(12, 5))
class_names = ['Fake(0)', 'Real(1)']
for ax, (name, (y_true, y_pred, _)) in zip(axes, results.items()):
    cm = confusion_matrix(y_true, y_pred)
    sns.heatmap(
        cm,
        annot=True,
        fmt='d',
        cmap='Blues',
        ax=ax,
        cbar=False,
        xticklabels=class_names,
        yticklabels=class_names)
    ax.set_title(f'Confusion Matrix - {name}', fontsize=12)
    ax.set_xlabel('Predicted', fontsize=10)
    ax.set_ylabel('Actual', fontsize=10)
plt.tight_layout()
plt.savefig('/kaggle/working/images/roberta-confusion_matrices.png
plt.show()

# Plot ROC-AUC curve plots
plt.figure(figsize=(10, 7))
for name, (y_true, _, y_proba) in results.items():
    fpr, tpr, _ = roc_curve(y_true, y_proba)
    roc_auc = auc(fpr, tpr)
    plt.plot(fpr, tpr, lw=2, label=f'{name} (AUC = {roc_auc:.3f})')
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('RoBERTa - ROC Curves Comparison')
plt.legend(loc="lower right")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('/kaggle/working/images/roberta-roc-auc.png', dpi=300,
plt.show()
```




RoBERTa - ROC Curves Comparison
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
RoBERTa-Imbalanced (AUC = 0.827)
RoBERTa-SMOTE (AUC = 0.843)
RoBERTa-RUS (AUC = 0.749)

In [138]:
```
# Print per-class and macro/weighted averages
for name, (y_true, y_pred, y_proba) in results.items():
    print(f"\n {name} Model:")
    print("-" * 40)
    print(classification_report(y_true, y_pred, target_names=['Fake
```

```
 RoBERTa-Imbalanced Model:
----------------------------------------
              precision    recall  f1-score   support

     Fake(0)     0.8000    0.3922    0.5263      1020
     Real(1)     0.8463    0.9716    0.9046      3515

    accuracy                         0.8412      4535
   macro avg     0.8232    0.6819    0.7155      4535
weighted avg     0.8359    0.8412    0.8195      4535


 RoBERTa-SMOTE Model:
----------------------------------------
              precision    recall  f1-score   support

     Fake(0)     0.5869    0.6686    0.6251      1020
     Real(1)     0.8998    0.8634    0.8812      3515

    accuracy                         0.8196      4535
   macro avg     0.7434    0.7660    0.7532      4535
weighted avg     0.8294    0.8196    0.8236      4535


 RoBERTa-RUS Model:
----------------------------------------
              precision    recall  f1-score   support

     Fake(0)     0.5307    0.5167    0.5236      1020
     Real(1)     0.8608    0.8674    0.8641      3515

    accuracy                         0.7885      4535
   macro avg     0.6958    0.6920    0.6939      4535
weighted avg     0.7866    0.7885    0.7875      4535
```

The findings justify that balancing the dataset used for finetuning pretrained transformers improves performance on the minority class.

- The RoBERTa transformer finetuned with SMOTE-balanced data ***RoBERTa-SMOTE*** scores a recall of **0.6686** for the minority class `Fake(0)` and achieves the highest macro-averaged recall (**0.7660**) indicating that the model performs best in terms of capturing both classes equally well.
- The RoBERTa transformer finetuned with RUS-balanced data ***RoBERTa-RUS*** scores a recall of **0.5167** for the minority class `Fake(0)` and a macro-averaged recall of **0.6920**.
- The RoBERTa transformer finetuned with imbalanced data ***RoBERTa-Imbalanced*** achieves the lowest recall of **0.3922** and a macro-averaged recall (**0.6819**).

In [152]:
```
# # Save Fine-tuned Models
# import os
# import shutil

# # Base directory to save models
# models_save_dir = "/kaggle/working/roberta_models_tuned"
# os.makedirs(models_save_dir, exist_ok=True)
# models_to_save = {
#     "roberta_imbalanced": model_imb,
#     "roberta_smote": model_smote,
#     "roberta_rus": model_rus}

# print("Saving fine-tuned RoBERTa models")

# for model_name, model in models_to_save.items():
#     model_path = os.path.join(models_save_dir, model_name)

#     # Remove old directory if exists
#     if os.path.exists(model_path):
#         shutil.rmtree(model_path)

#     # Save model and tokenizer
#     model.save_pretrained(model_path)
#     roberta_tokenizer.save_pretrained(model_path)
#     print(f"Model saved to: {model_path}")
```

4. Evaluation and Model Interpretability

Objective 4: Interpret best performing ML model using the LIME library

Best Performing Model

MODEL	MACRO-AVERAGED RECALL	RECALL (Fake News)	AUC
Logistic Regression-Imbalanced	0.6774	0.3980	0.816
Logistic Regression-SMOTE	0.7145	0.7029	0.815
Logistic Regression-RUS	0.7368	0.7000	0.815
XGBoost-Imbalanced	0.6854	0.4245	0.816
XGBoost-SMOTE	0.7248	0.5627	0.819
XGBoost-RUS	0.7427	0.7088	0.822
LSTM-Imbalanced	0.6344	0.2990	0.797
LSTM-SMOTE	0.7104	0.5784	0.792
LSTM-RUS	0.6948	0.6539	0.759
Finetuned RoBERTa-Imbalanced	0.6819	0.3922	0.827
Finetuned RoBERTa-SMOTE	0.7660	0.6686	0.843
Finetuned RoBERTa-RUS	0.6920	0.5167	0.749

The Finetuned RoBERTa model using SMOTE-balanced dataset is chosen as the better performing model because it achieves the highest scores for **Macro-averaged recall** (0.7660). This is corroborated by the confusion matrix, which reports the highest true positives for accurate fake news predictions (682). The ROC-AUC curve plots reinforces this decision since with the finetuned transformer with SMOTE-balanced data achieves the highest AUC (0.843). RoBERTa's contextual understanding from pretraining and fine-tuning with balanced data enhances its robustness, and generalizability.

Interpret the Finetuned RoBERTa-base Model

We utilize the LIME (Local Interpretable Model-agnostic Explanations) library to interpret the predictions of a fine-tuned RoBERTa model, which classifies news articles as either class 0 (fake) or class 1 (real). We load the pre-trained RoBERTa model and its tokenizer from a specified path, setting the model to evaluation mode to disable gradient computation. A predictor function tokenizes input texts, processes them through the model, and applies a softmax function to convert logits into probability scores for the two classes.

The LIME explainer is initialized with class names 'real' and 'fake', enabling local interpretation of a selected news article (e.g., `data.iloc[99, 10]`). It generates explanations by perturbing the text and evaluating the model's predictions on these variations, using 500 samples and highlighting the top 10 features. The results are visualized as a 1x2 horizontal bar plot, with one subplot per class. Bars represent feature weights:

- Green for positive influence (supporting the class).
- Red for negative influence.

The bars are sorted by absolute weight to emphasize the most impactful words. Leveraging the LIME library provides insight into which terms drive the model's classification, enhancing interpretability of the fake/real news prediction.

In [150]:
```
# Preview a random sample
data.iloc[99, 10]
```

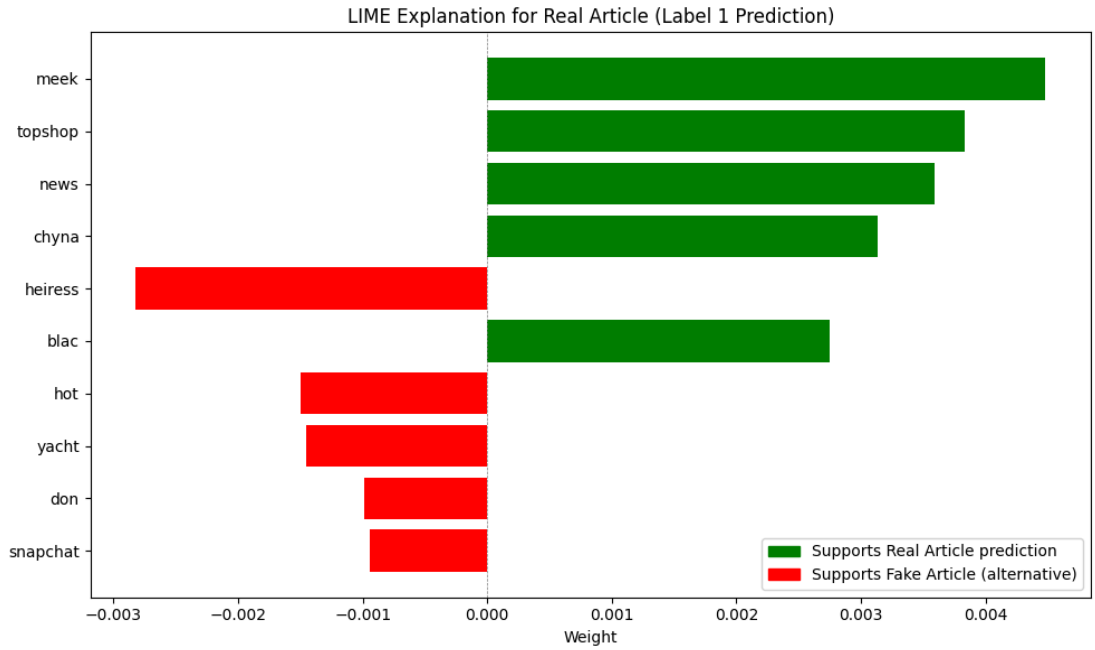
Out[150]:
```
'blac chyna cozy hot felon jeremy meek pic look like blac chyna h
ot felon jeremy meek work chyna share picture posing arm snapchat
wednesday night show tattoo chyna don skintight lace orange dress
meek keep casual camouflage print shirt rip jean news early snapc
hat post chyna appear photo shoot lead speculate possibly star ca
mpaign meek model attractive mugshot go viral headline summer rel
ationship topshop heiress chloe green meek photograph kiss green
luxury yacht coast turkey july married wife melissa time shortly
picture surface meek file separation melissa year marriage news m
eek green shy pack pda watch video'
```

In [151]:

```
# Load the fine-tuned model and tokenizer
model_path = "/kaggle/input/roberta_smote/transformers/default/1/rc
tokenizer = RobertaTokenizerFast.from_pretrained(model_path)
model = RobertaForSequenceClassification.from_pretrained(model_path
model.eval()

# Define the LIME-compatible predictor function
def predictor(texts):
    encodings = tokenizer(texts, truncation=True, padding=True, ret
    with torch.no_grad():
        outputs = model(**encodings)
    probas = torch.nn.functional.softmax(outputs.logits, dim=1).nur
    return probas

# LIME explainer instance
class_names = ['Fake Article', 'Real Article']
explainer = LimeTextExplainer(class_names=class_names)

# Choose an article to explain
sample_text = data.iloc[99, 10]

# Get prediction
probas = predictor([sample_text])
predicted_label = int(np.argmax(probas))  # 0 or 1
predicted_class_name = class_names[predicted_label]

# Generate explanation for the predicted label
explanation = explainer.explain_instance(
    sample_text,
    classifier_fn=predictor,
    num_features=10,
    num_samples=500,
    labels=[predicted_label])

# Extract top-10 features for the predicted label
features = explanation.as_list(label=predicted_label)
features_sorted = sorted(features, key=lambda x: abs(x[1]), reverse
features, weights = zip(*features_sorted[:10])
print(f"Predicted Class: {predicted_class_name} (Label {predicted_l

# Plot a horizontal bar plot to visualize top-10 important features
y_pos = np.arange(len(features))
colors = ['green' if w > 0 else 'red' for w in weights]
plt.figure(figsize=(10, 6))
plt.barh(y_pos, weights, align='center', color=colors)
plt.yticks(y_pos, features)
plt.gca().invert_yaxis()
plt.xlabel('Weight')
plt.axvline(x=0, color='gray', linestyle='--', linewidth=0.5)
plt.title(f"LIME Explanation for {predicted_class_name} (Label {pre

# Define legend patches
if predicted_label == 0:
    green_label = "Supports Fake Article prediction"
    red_label = "Supports Real Article (alternative)"
else:
    green_label = "Supports Real Article prediction"
    red_label = "Supports Fake Article (alternative)"
green_patch = mpatches.Patch(color='green', label=green_label)
red_patch = mpatches.Patch(color='red', label=red_label)
plt.legend(handles=[green_patch, red_patch], loc='lower right')
```

```
plt.tight_layout()
plt.savefig('./images/lime_explanation_barplot.png', dpi=300, bbox_
plt.show()
```

```
Predicted Class: Real Article (Label 1)
```



5. Deployment

Objective 5: Deploy selected model using FastAPI and Streamlit

Thresholding Strategy

An asymmetric thresholding strategy was adopted to further improve the model's ***recall*** at the expense of a slight drop in ***precision*** as follows:

- The threshold for predicting Fake Articles was lowered to **0.45**.
- The threshold for predicting Real Articles was raised to **0.60**.
- Probabilities between (**0.40 to 0.45**) for ***Label 0*** and (**0.55 to 0.60**) for ***Label 1*** were flagged for human review.

Containerization Strategy

We leveraged FASTAPI, Streamlit, and Docker to deploy the fine-tuned RoBERTa Transformer Model for fake news detection.

- **Back-end :** The FastAPI app (***main.py***) is responsible for model inference. It loads the saved roberta_model, incorporates the SpaCy text preprocessing pipeline, scrapes article text from a provided URL using BeautifulSoup, predicts whether it is fake or real, and generates LIME explanation results.
- **Front-end :** The Streamlit app (***app.py***) provides an interactive frontend web interface that prompts the user to enter a news article URL for prediction. It sends the URL to the FastAPI endpoint (/predict/) via localhost, displays the article title, prediction label, and generates a LIME explanation plot.

- **Docker :** Used to containerize the front-end and the back-end using two Dockerfiles (one for each).
  - The required Python dependencies for the front-end include: streamlit, requests, and matplotlib.
  - The required Python dependencies for the back-end include: fastapi, uvicorn, transformers, torch, numpy, lime, pydantic, requests, beautifulsoup4, spacy, and lxml.
  - The docker-compose.yml exposes ports: **8000:8000** to the back-end and ports

6. Conclusion, Recommendations, and Next Steps

6.1 Conclusion

This project successfully developed a robust pipeline for fake news detection. It leveraged advanced NLP techniques and multiple machine learning approaches. The dataset (15,116 news articles) was meticulously processed through feature engineering (domain extraction, text length analysis, preprocessing) and exploratory analysis. The ***macro-averaged recall*** metric was selected as the success criteria because the project aims to catch fake news articles but the training dataset is imbalanced. The Key objectives were addressed by:

- Utilizing the spaCy library to preprocess text data and perform EDA to unveil underlying characteristics among fake and real news articles.
- Building a baseline **Logistic Regression** model to establish initial performance benchmarks.
- Building an ensemble model (**XGBoost**), a neural network (**LSTM**), and finetuning the **RoBERTa-base** transformer.
- Evaluating the models' respective performance on test set and interpreting the alternative with the highest ***macro-averaged recall*** using LIME.
- Deploying the selected ML model (**Finetuned RoBERTa-base transformer with SMOTE-balanced dataset**) using FastAPI, Streamlit, and Docker.

6.2 Recommendations

- Incoporate more training data expecially class 0 (Fake News) entries to improve recall.
- Leverage pretrained embeddings when finetuning the RoBERTa transformer to reduce training time.
- Experiment finetuning other NLP transformers such as DistilBERT and DeBERTa.

6.3 Next Steps

1. Deploy the containerized finetuned RoBERTa model via AWS SageMaker for low-latency inference.
2. Develop browser extensions and social media plugins using the model’s API for real-time credibility scoring.
3. Incorporate automation frameworks to flag low-confidence predictions for human review.